

Low Speed Flow Past a High Aspect Ratio Wing Lab

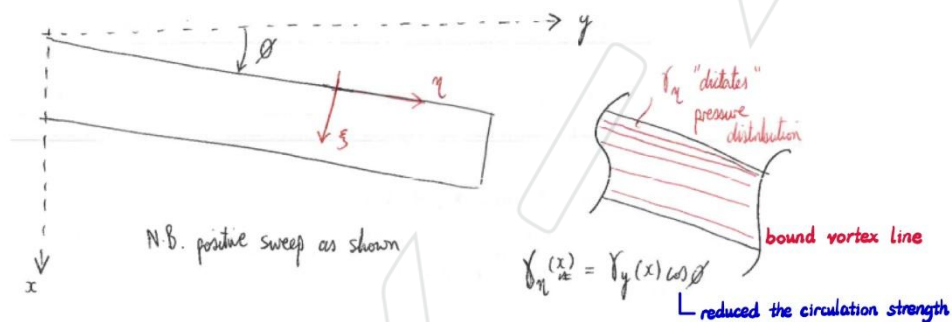
Pre-Lab Notes

You should be familiar with Prof. Buxton's lecture [Chapters 1, 2, 3, 5.1 and 5.2](#) before reading this note

Swept Wing Lecture

Infinite swept wing

Only 3D influences arise due to sweep alone

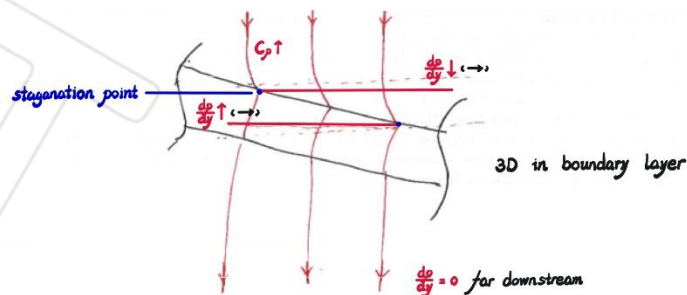


Bounded vortex in the section of the wing

1. Delay of M_{crit} in high-Mach-number subsonic flight, as the wing can remain subcritical
 - Shock wave formation
 - Flow becomes compressible
 - Fly beyond M_{crit} might reach M_{DD} where wave drag increases dramatically

* Adding sweep also could result in thinner wing, which reduces 'acceleration' of flow on wing surface

2. Velocity perturbation changes to $u_{\xi} = u_{\infty} \cos \phi$ and $u_{\eta} = u_{\infty} \sin \phi$

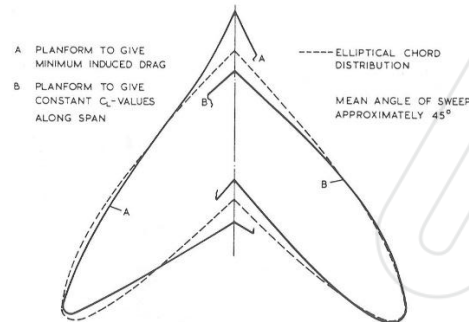


Streamline curved over the wing (non-zero pressure gradient being created)

* For wing of large AR , the conditions of infinite sweep almost meet at the half-wing

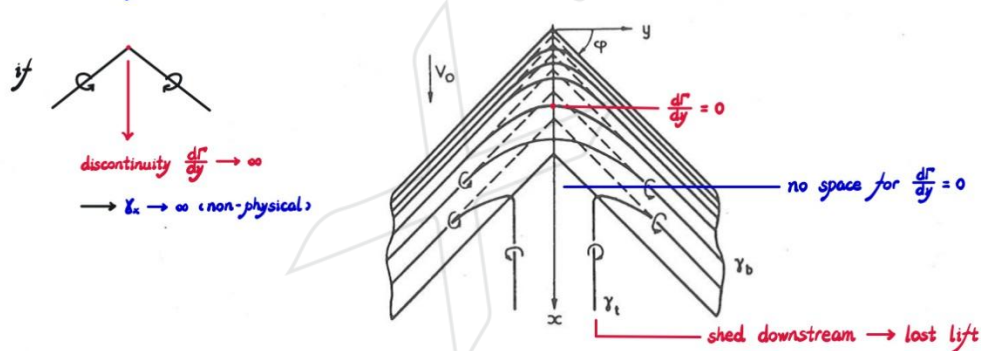
Finite aspect ratio

When $AR \geq 4$, we say it is large AR . When $AR \leq 3$, the effects at wing tip and root merge.



Platforms of minimum induced drag, constant lift and elliptical chord are no longer overlaid
 – lifting line theory becomes invalid

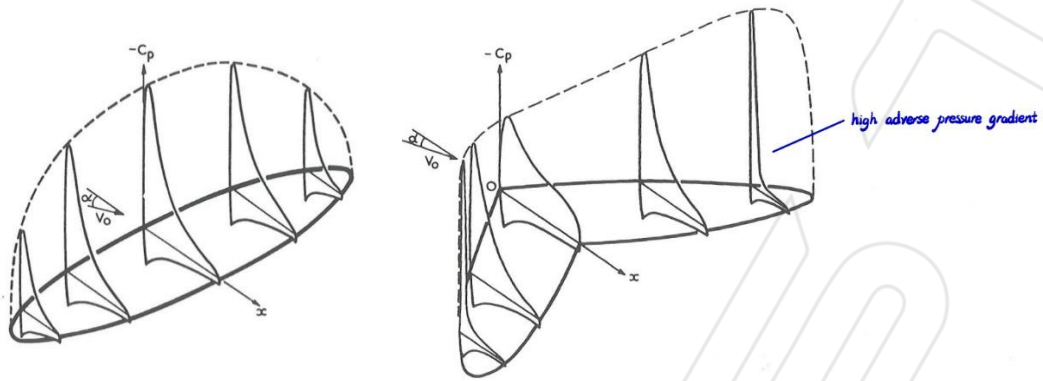
$$\gamma_x = \pm \frac{d\Gamma}{dy}$$



Vortex pattern near the center of a sweptback wing - Shed Γ at trailing edge

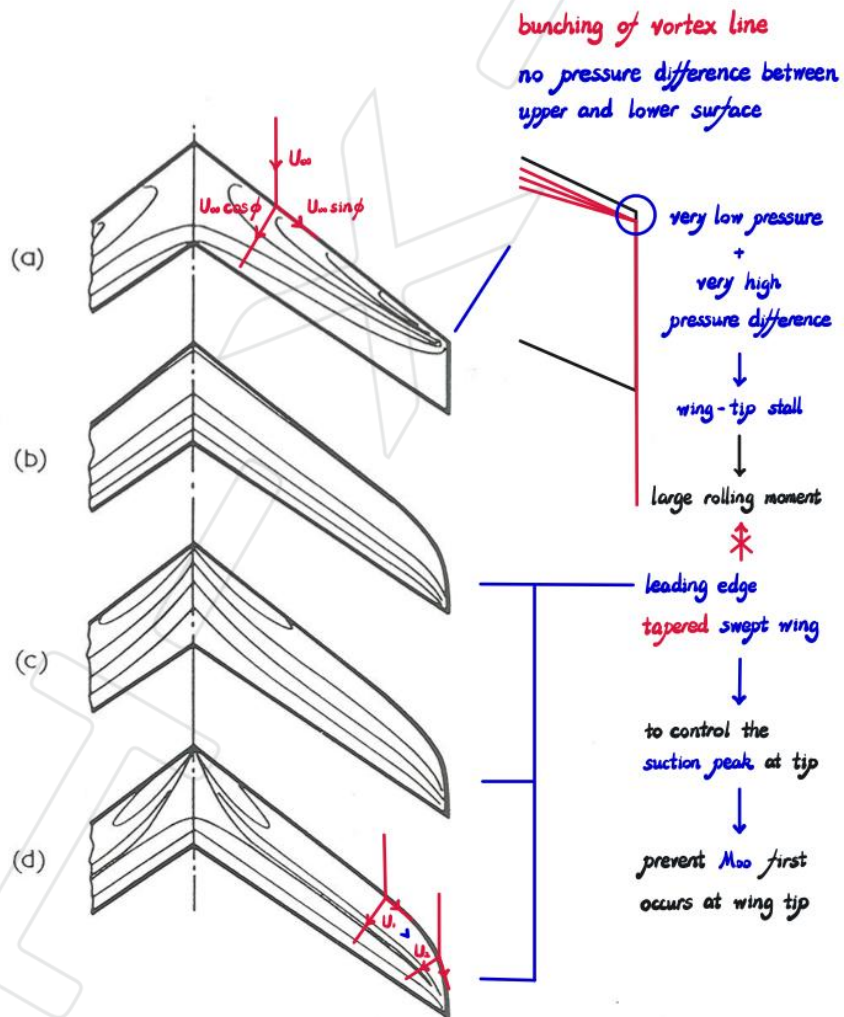
1. At root
 - Shed Γ at trailing edge - overall loss of lift
 - Aerodynamic centre moves aft
2. At tip
 - Increase suction peak pressure gradient
 - Aerodynamic centre moves forward
 - Potential tip stall due to larger adverse pressure gradient (also because spanwise flow that pushes the boundary layer outboard) – results huge rolling moment
 - Earlier sonic condition at tip than root due to addition of transverse flow

* Taper leading edge to overcome the last two issues



Pressure distribution over flat and swept back wing with the large aspect ratio

- Note how the effect of shed vorticity at the centre line is to push the pressure peak downstream, while the effect of the tip trailing vorticity is to push the pressure peak upstream. As a result, the pressure drag of the swept wing is highest at the centre.



Effect of leading edge tapered swept wing

A. Uniform taper

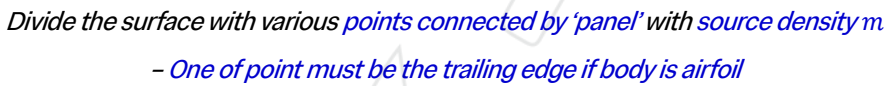
- Reduced lift at the tip - smaller chord at the tip
- Reduced risk of tip stall - aerodynamic load is shifted more towards the root
- More elliptical lift distribution
- Counteract the adverse effects of spanwise flow

B. Linear Twist (Washout)

- The root section of the wing will reach its stall angle before the tip - reducing the effective angle of attack at the wingtip compared to the root
- Potential tip stall due to larger adverse pressure gradient (also because the spanwise flow that pushes the boundary layer outboard) – results in a huge rolling moment
- Counteract the adverse effects of spanwise flow

*Will tip stall first in our experiment?

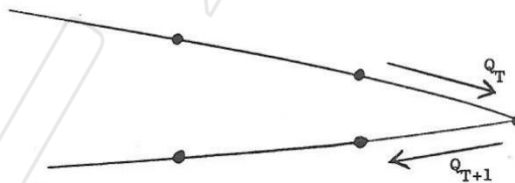
Method



θ_p measured anticlockwise from x axis to inside surface of element.



1. Find **velocity** components at the control point of the p^{th} element due to the q^{th} element
 - Find v_{xq} and v_{yq} as function of m_q
 - Find **normal velocity due to the q^{th} element** at p^{th} element $v_n(p, q) = A_{p,q} m_q$
 - Find **tangential velocity due to the q^{th} element** at p^{th} element $v_t(p, q) = B_{p,q} m_q$
2. Find **velocity** components at p^{th} element due to free stream and AoA
 - Find **free stream normal velocity** at p^{th} element $U_n(p) = U_\infty \sin(\alpha - \theta_p)$
 - Find **free stream tangential velocity** at p^{th} element $U_t(p) = U_\infty \cos(\alpha - \theta_p)$
3. Compute **source density** at p^{th} element
 - **Normal velocity** at p^{th} element $\sum_{q=1}^N v_n(p, q) + U_n(p) = 0$ (linearized boundary condition)
4. Compute **tangential velocity components** at p^{th} element
 - **Tangential velocity** at p^{th} element $Q_p = \sum_{q=1}^N v_t(p, q) + U_t(p)$
5. Find **additional source density** due to the application of the **unit circulation** $\Gamma = U_\infty c$
 - **Normal velocity** at p^{th} element $\sum_{q=1}^N v_{n\Gamma}(p, q) = U_{n\Gamma}(p)$
6. Find number k of unit circulation that **fulfills Kutta condition at the trailing edge**
 - **Kutta condition is** $Q_T + Q_{T+1} = \Delta Q_{t.e} = 0$
 - Therefore, $k \Delta Q_{\Gamma t.e} + \Delta Q_{t.e} = 0$



7. Find the **surface velocity** at p^{th} element **by superposition**
 - **Normal velocity** at p^{th} element $\widehat{Q}_p = Q_p + k Q_{p\Gamma}$
 - **Pressure coefficient** at p^{th} element $C_p(p) = 1 - \frac{\widehat{Q}_p^2}{U_\infty^2}$
8. Find lift coefficient
 - **Lift coefficient** $C_L = 2k$

Pros and cons of using panel code estimate swept wing aerodynamics

1. **Linear boundary problems** can be computed when **viscous effects** are either unimportant or can be dealt with in a simpler way such as by the addition of a circulation

Pros:

- **Simplicity**
- **Faster Computation** by avoiding the computational complexity of full Navier-Stokes or boundary layer models

Cons:

- Cannot accurately capture **flow separation** or **stall** behaviour due to **boundary layer detachment**
- **Poor drag estimation** - **pressure drag** and **skin friction drag** are not directly accounted for by inviscid panel codes
- **Complex flow interaction** on the swept wing, involving spanwise flow, adverse pressure gradients, and tip vortices, panel code cannot fully capture the complex interplay of **3D viscous forces**

2. Lifting problem in 3D

Cons:

- **Very lengthy iteration processes** are needed to obtain the correct position for shed trailing vortex wake

* Test the accuracy of the panel code by changing the **number of elements** around the airfoil (A.M.O Smith suggests 3D bodies 500-1000 panels)

** **Curved elements** can be used for 3D bodies for more algebra but fewer elements

cd "C:\Users\GH-X-ST\OneDrive - Imperial College London\Year 3\Aerodynamics 3\A - Low Speed Flow Past a High Aspect Ratio Wing Lab\01 - Lab Files\panel_code"

Measure-Command { .\combo.exe mesh.dat auxiliary.dat stream.dat sect.inp }

Effect of resolution

1. Chord-wise discretisation

- **Finer discretization leading a more accurate lift calculation**, as described the shape of the airfoil

2. Span-wise discretisation

- **Diverging occurs at the wing tip** when the number of span-wise discretisation is smaller or equal to the number of section (3 section in this experiment)

Experiment

Procedure

1. Read Pressure

- Set wing to $\alpha = 0^\circ$
- Record [manometer readings](#) to data.xlsx
- Repeat by set wing to $\alpha = 6^\circ$ and $\alpha = 13^\circ$

*Change AoA gradually to [avoid the stepper motor stall](#) (when the motor does not have enough torque to overcome the load when accelerating/decelerating too fast)

2. Stall the wing

- Set wing back to $\alpha = 0^\circ$
- Increase incidence
- [Record incidence at a stall](#) by observing the [wool tufts](#) and [pressure distribution](#)
- Make sketches of the qualitative shape of the pressure distribution for each of section

*Wait some time to make stall progress throughout the wing before recording

**Try to record different threshold value of AoA for tip and root stall

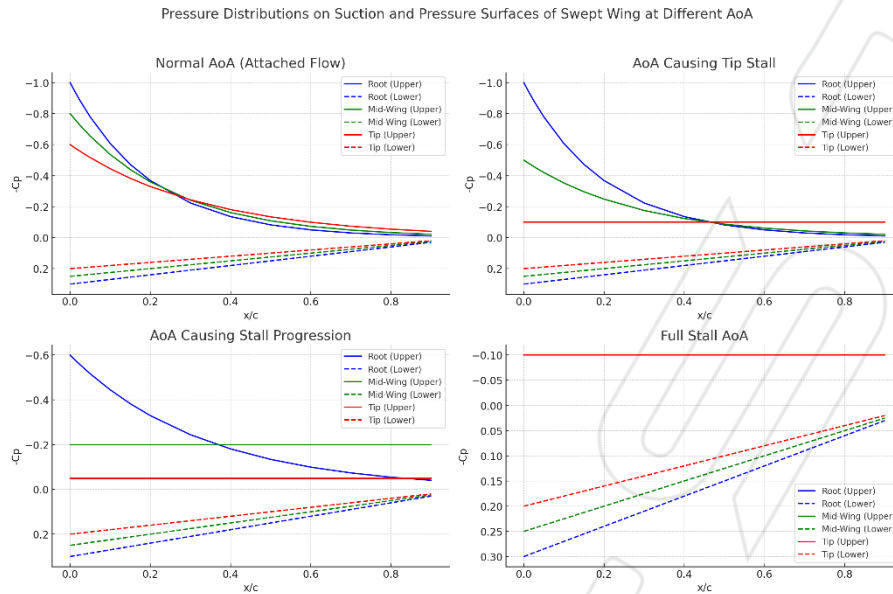
*** Stall angle for infinite wing in [3] should be 17 degrees

A. Wool tufts

- Post stall - [lie smoothly](#) along the surface of the wing while [pointing slightly outboard](#) due to the spanwise component of flow
- Tip Stall (which is expected for swept wing) Initiation - wool tufts [near the wingtip](#) will flutter and oscillate
- Separation progresses inboard
- Full Stall - wool tufts across most or all of the wing will be [flapping wildly](#)

*[Record videos](#) of the wool tufts at above experiments and add [screenshots](#) to the PowerPoint

B. Pressure distribution at different AoA



Expected pressure distribution at different AoA

Conclusion

1. The influence of sweep and infinite aspect ratio on the sectional lift-coefficient slope of a swept wing, is panel code a good approximation?

- Provide plots in MATLAB for swept wing
- Record [manometer readings](#) to data.xlsx
- Repeat by set wing to $\alpha = 6^\circ$ and $\alpha = 13^\circ$

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